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# A Self-Adaptive Digital Twin Architecture for Dynamic Resource Management

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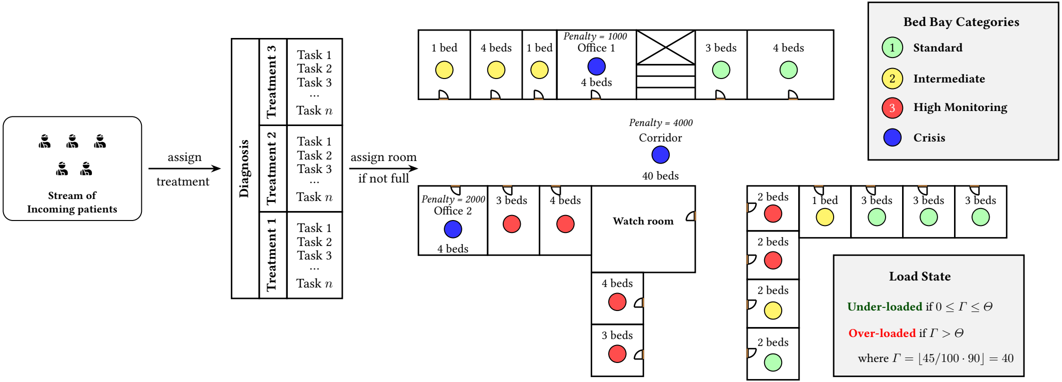
SEAMS 2026

[+]University of Oslo [*]University of Durham

April 13, 2026

# Contents

# Why do we need it?



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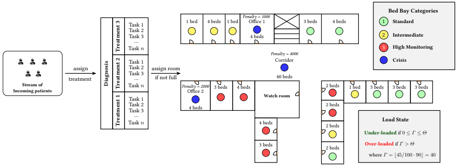


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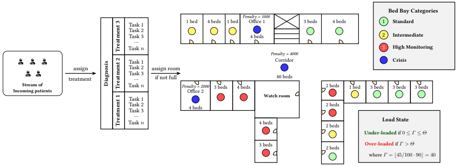


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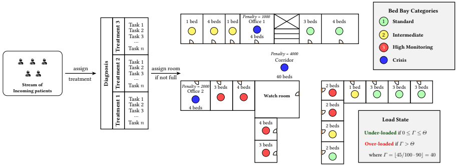


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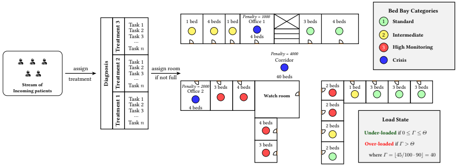


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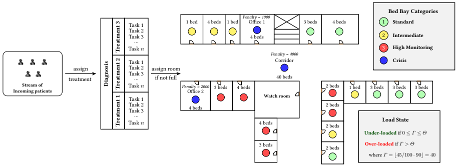


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- We need runtime adaptation with trustworthy model alignment

# Baseline: BedreFlyt and the gap

- **BedreFlyt** is a hospital-ward digital twin that combines an evolving semantic model, simulation-driven forecasting, and optimisation to recommend bed-bay allocations to staff

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- Human-in-the-loop actions can diverge from DT suggestions
- **Main gap**: the original ward layout is treated as fixed

# What is the goal

**Goal**: enable the digital twin to reconfigure ward capacity at runtime (open/close spare rooms) in response to changing patient demand, while staying aligned with the real ward and keeping a human in the loop.

- A self-adaptive architecture for resource management (BedreFlyt use case) that combines

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- Human-in-the-loop supervision over the adaptation loop

# Contents

# What is a Digital Twin

Digital twins are, as defined by NASEM

## NASEM Definition of DT (2024)

A digital twin is a set of virtual information constructs that mimics the structure, context, and behaviour of a natural, engineered, or **social system** (or system-of-systems), is dynamically updated with data from its physical twin, has a predictive capability, and **informs decisions** that realise value. The bidirectional interaction between the virtual and the physical is central to the digital twin[a].

[a] National Academies of Sciences, Engineering, and Medicine (NASEM) (2024): Foundational Research Gaps and Future Directions for Digital Twins. The National Academies Press

# Digital Twins components

**Insight:**
visualisation, dashboards, analyses

↕

**Information:**
organised data

↕

**Data:**
observations, parameters, domain model

## Capabilities for different insights:

- **Descriptive:** Insight into the past ("what happened")

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- **Reactive:** Automated decision making

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**Software layer**
monitor, analyse, plan, execute

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domain knowledge, asset and twin model

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▪ DT must adapt both structurally and behaviourally

▪ States of the DT cycles over time and they need to be captured

# Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

smolang.org

# Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

## SMOL: Semantic Model Object Language

- Smalll OO programming system
- Ontology reasoners allow querying the KB
- Used to create the digital model

smolang.org

# Twinning in SMOL

Programming support for twins with a behavioural and a structural layer

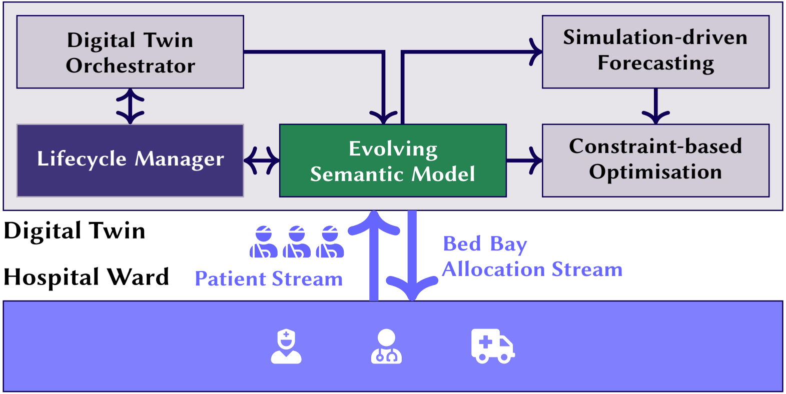
smolang.org

## behavioural twins in SMOL

- SMOL can encapsulate (simulation) models based on the FMI standard
- Can automatically adapt the object states in the KB at runtime through semantic lifting

# Contents

# Solution: DynResDT (Self-Adaptive DT Architecture)



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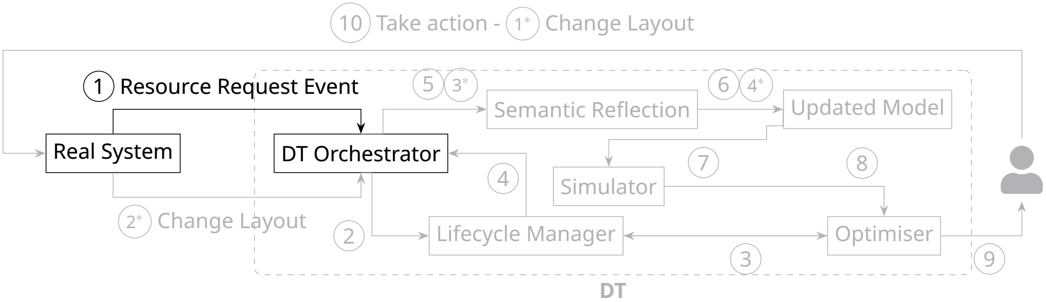

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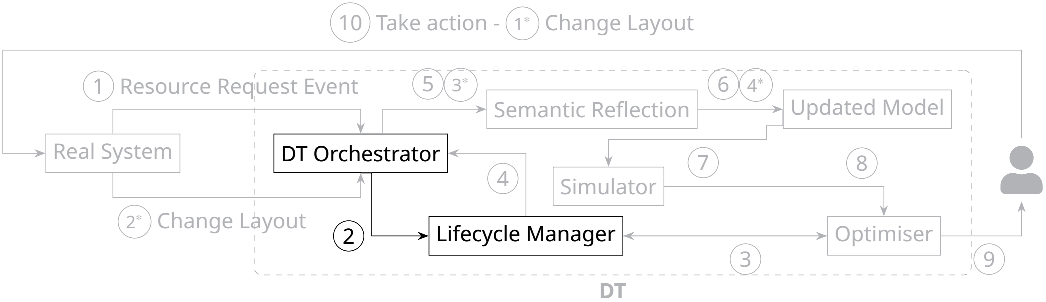

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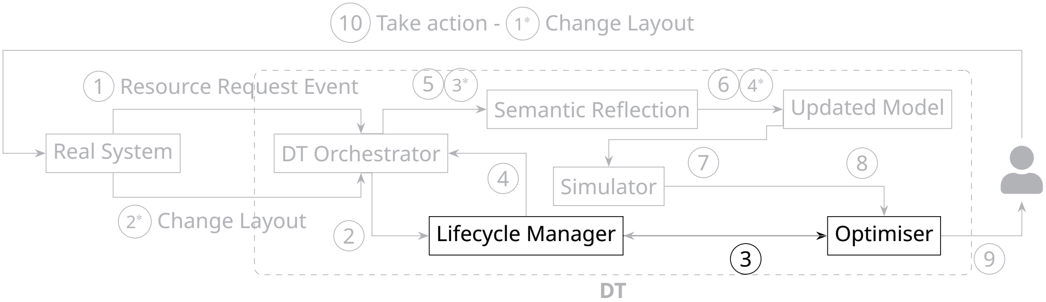

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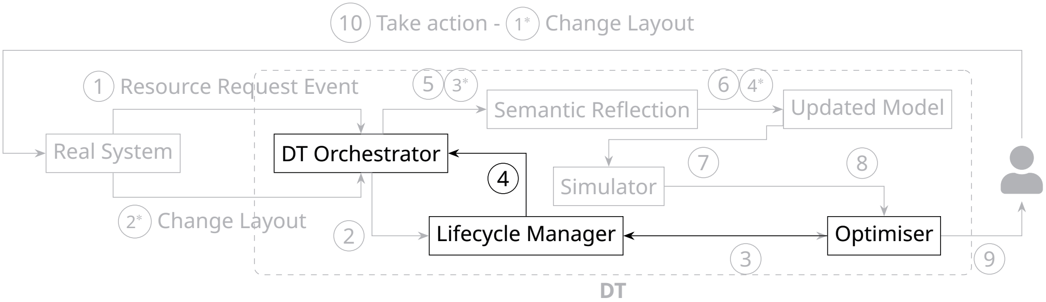

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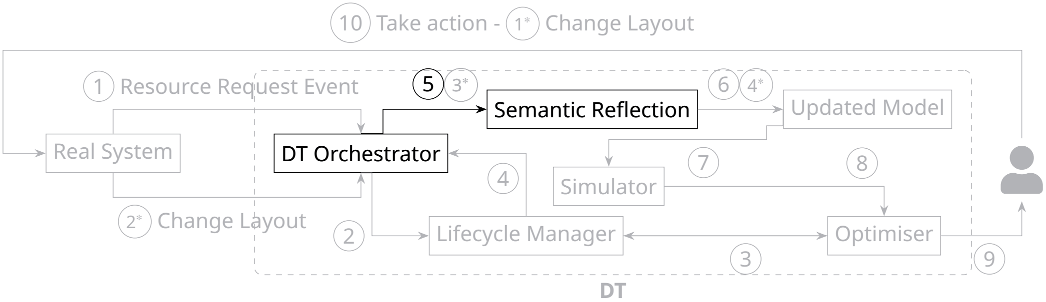

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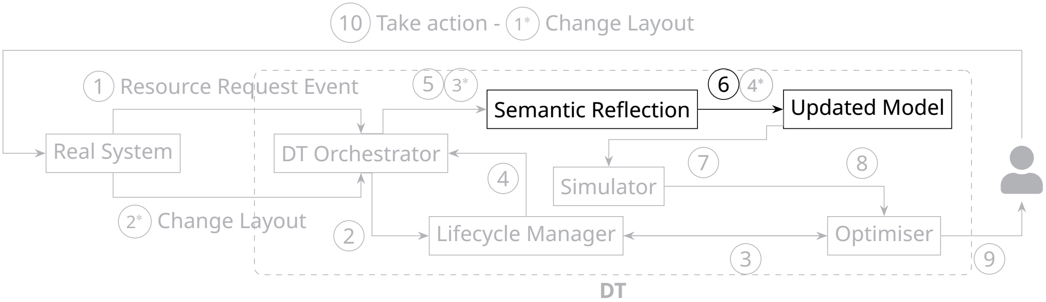

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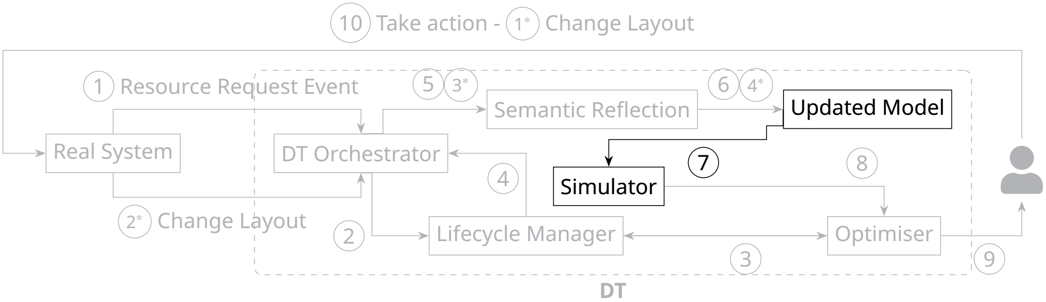

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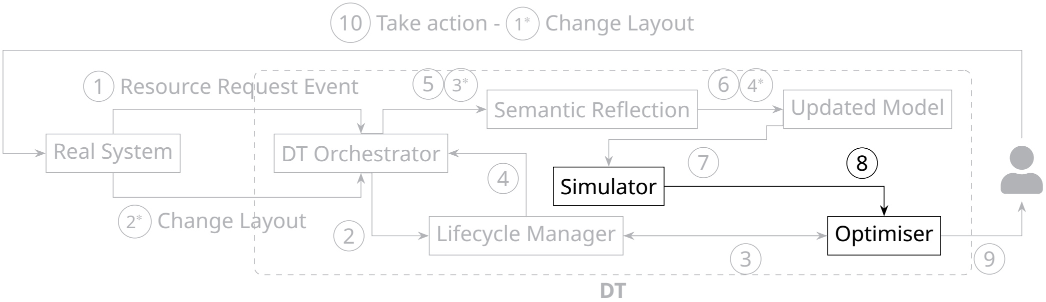

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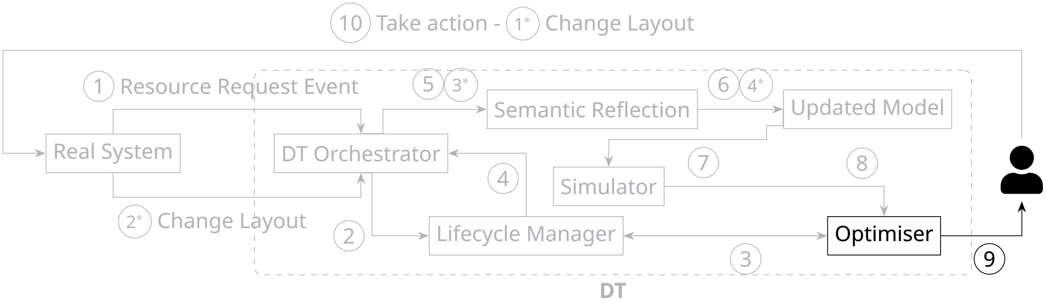

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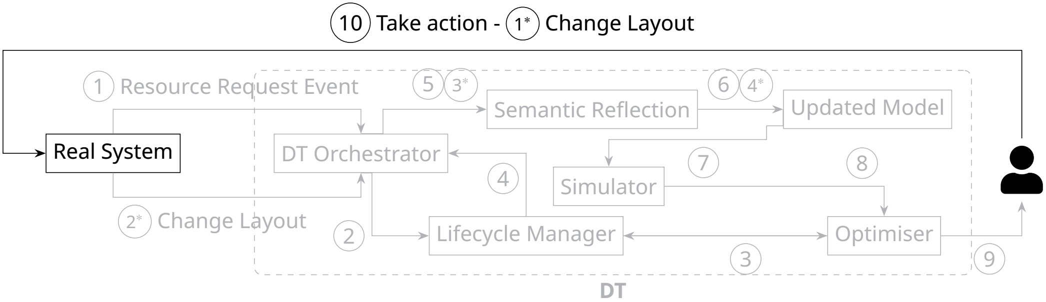

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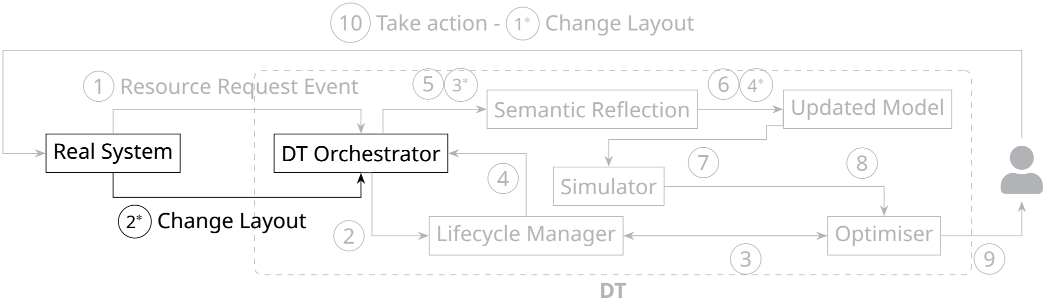

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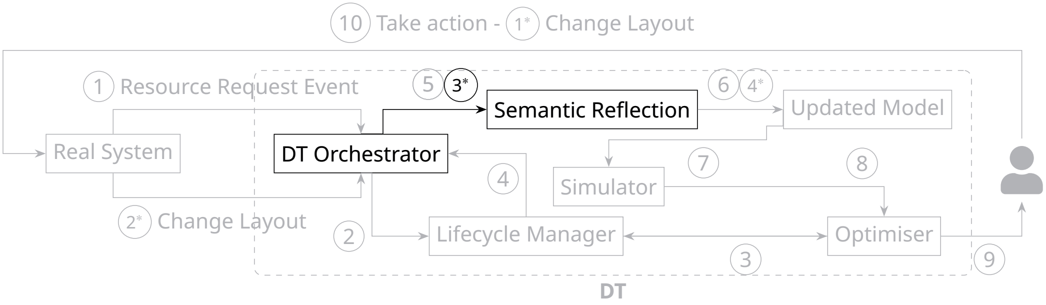

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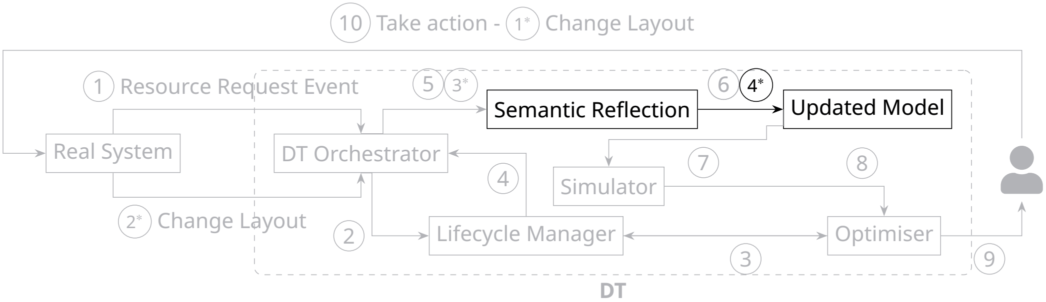

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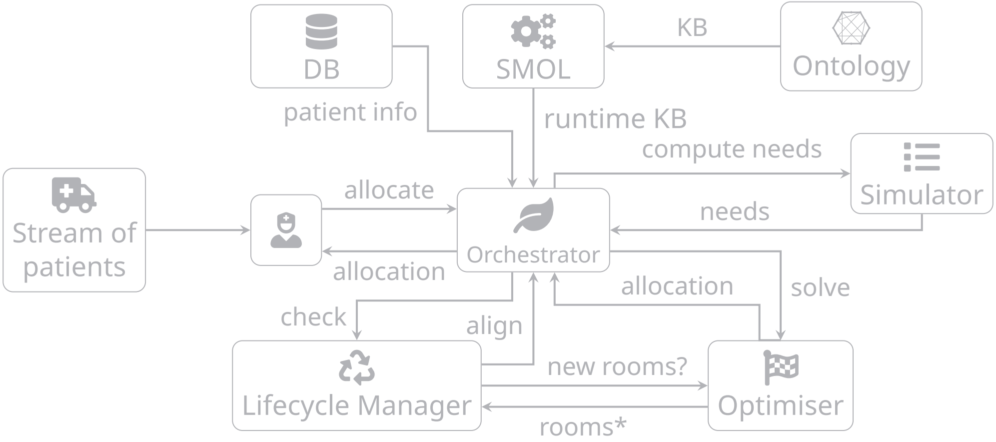
- Semantic reflection re-aligns the runtime model after structural changes
- This enables supervised autonomy: controlled self-adaptation instead of static decision support

# Allocation of patients



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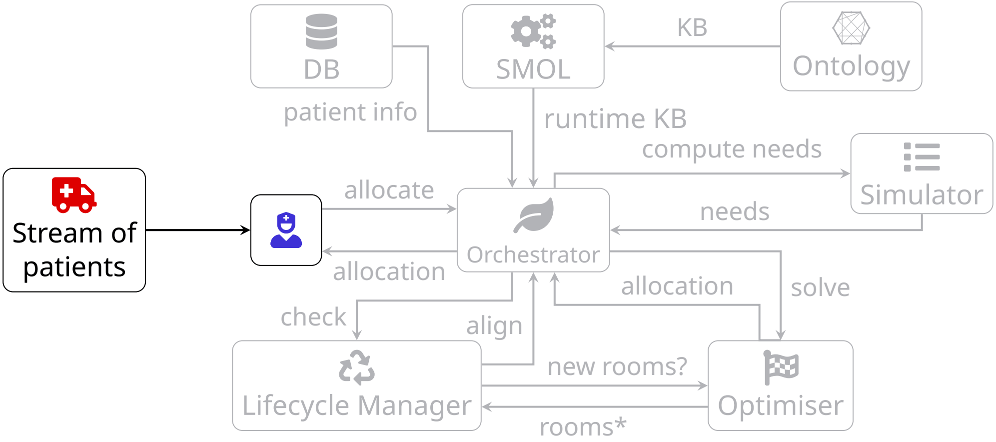

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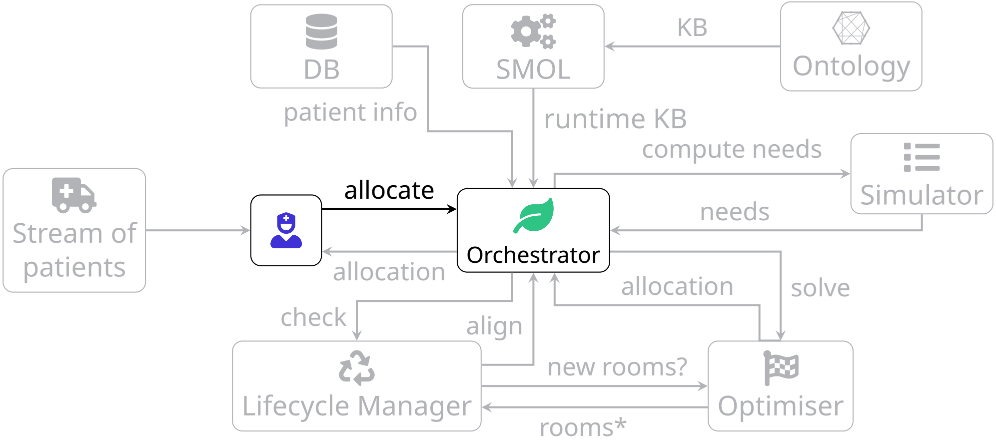

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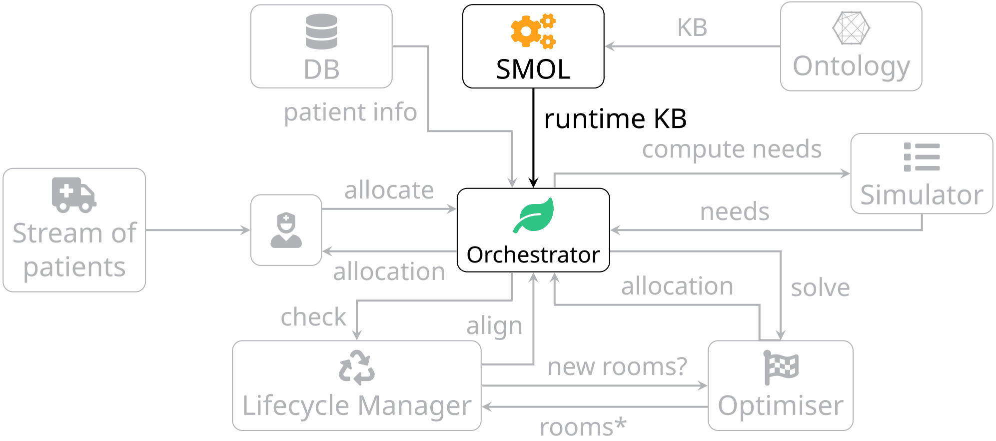

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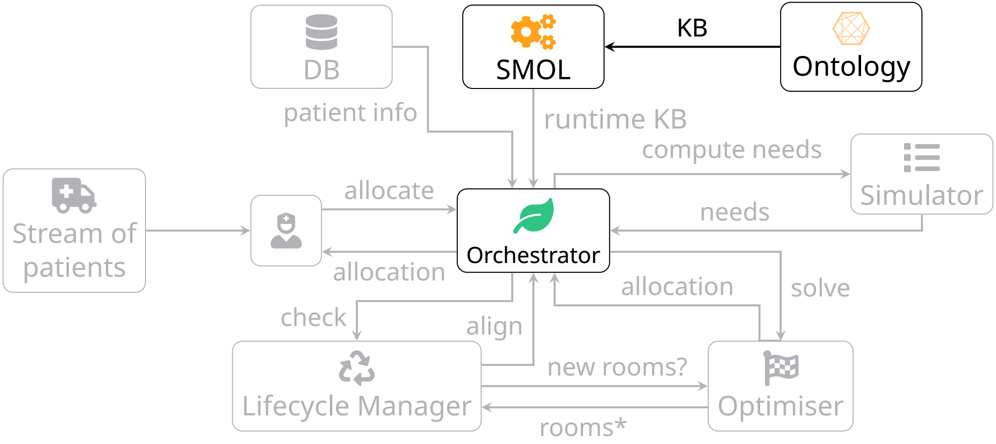

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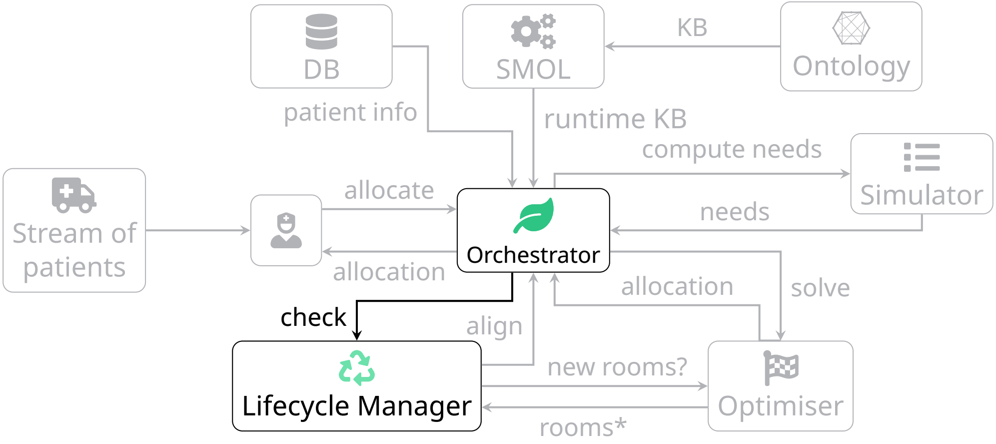

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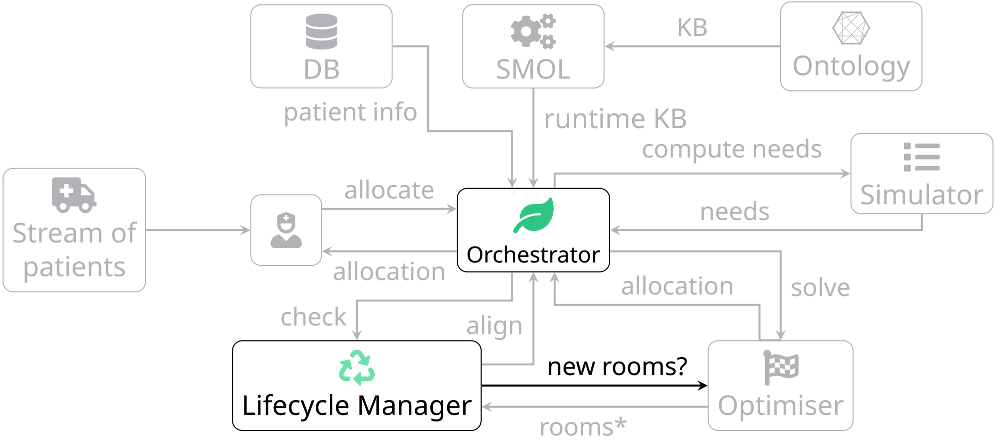

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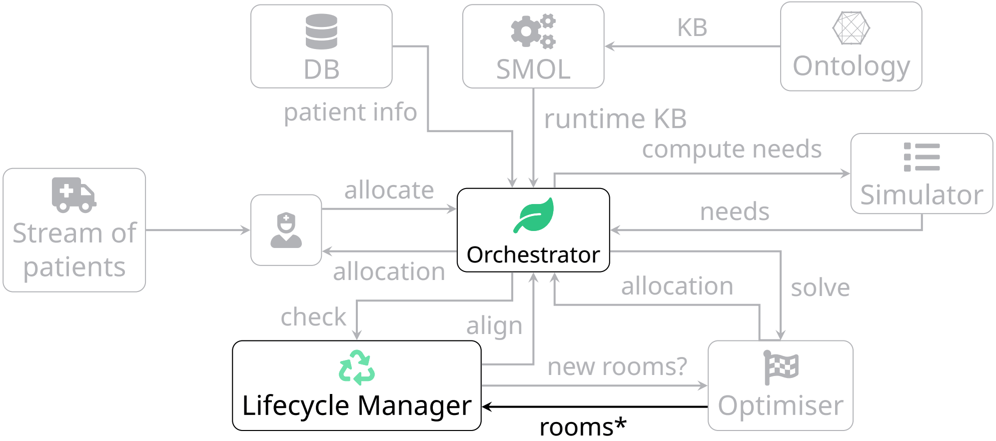

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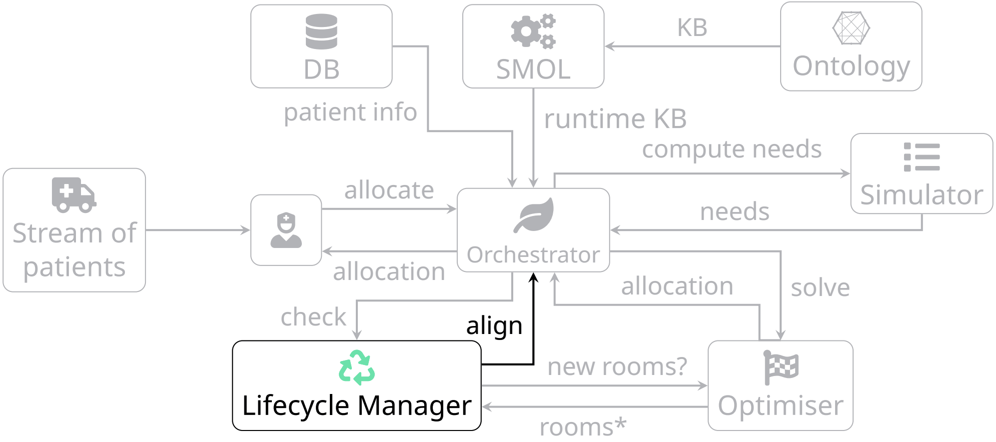

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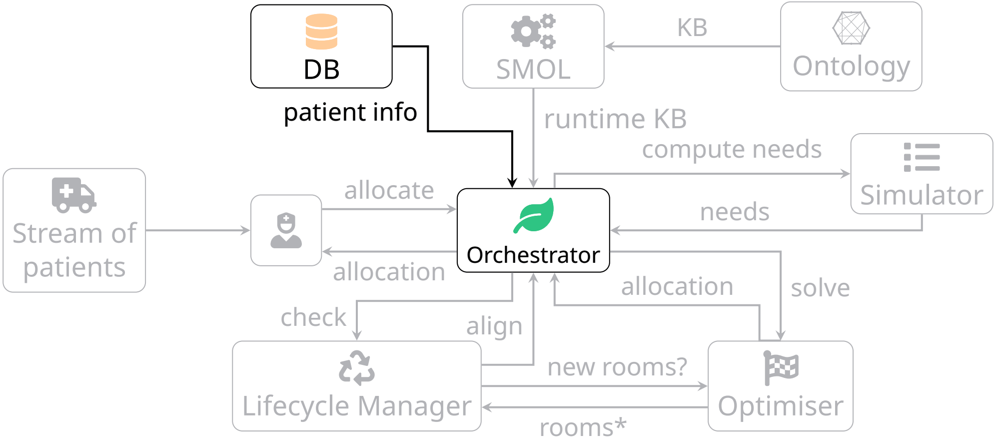

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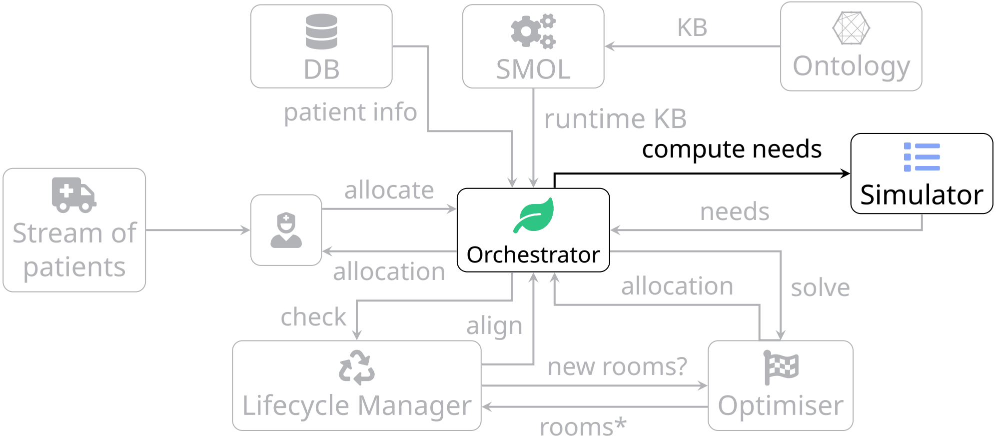

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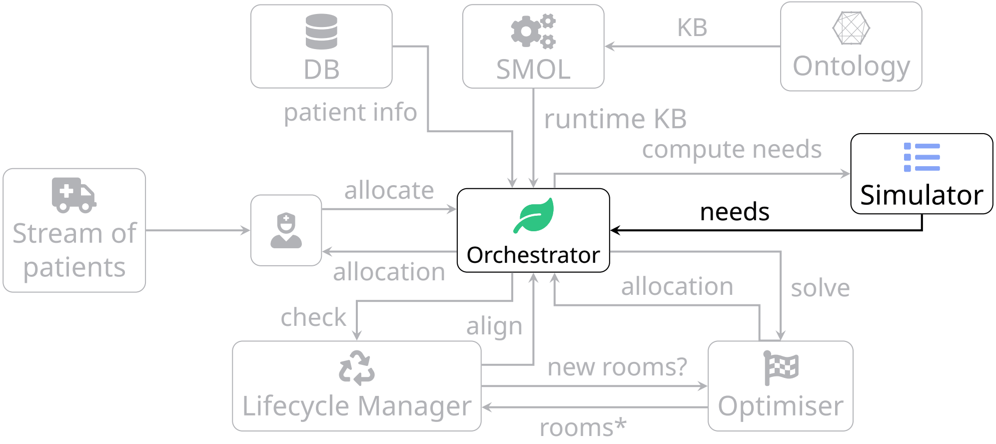

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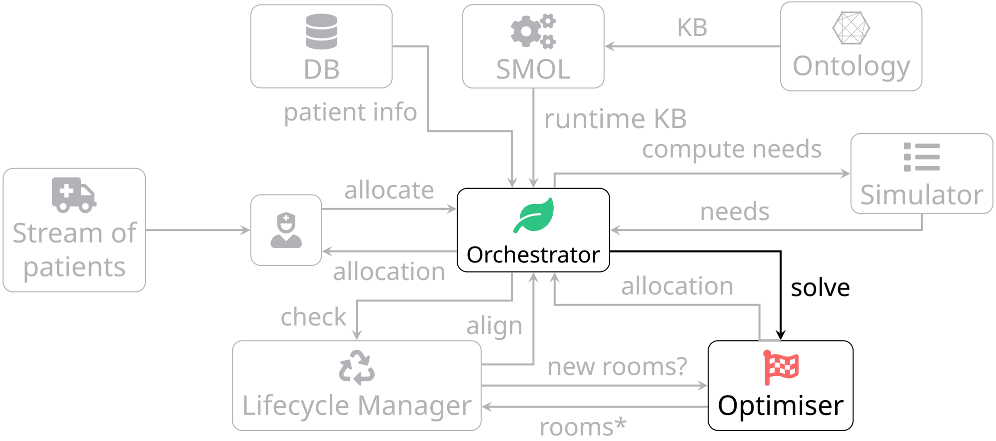

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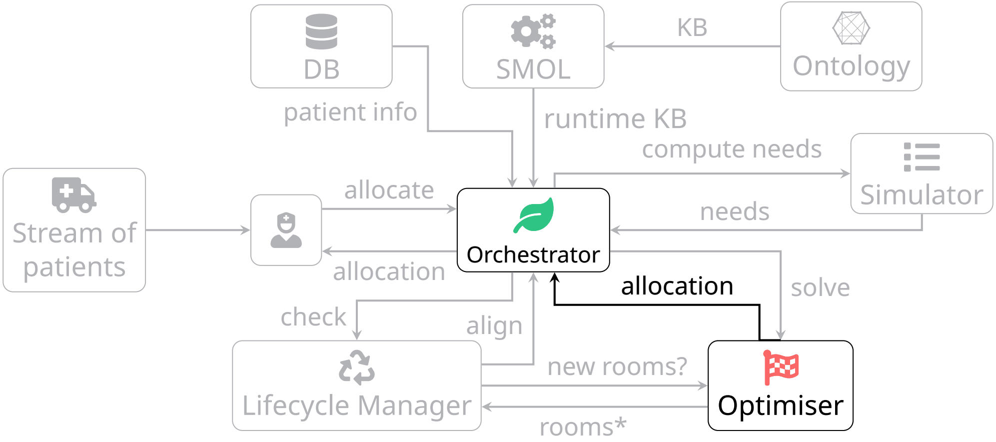

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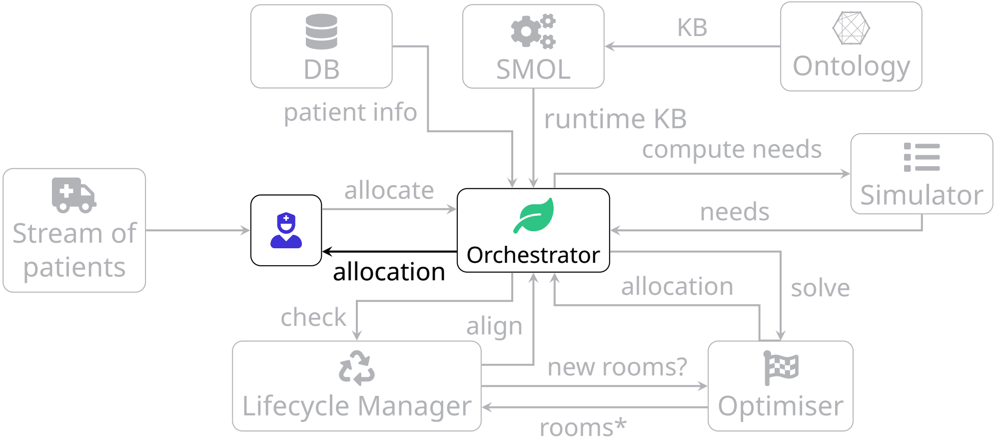

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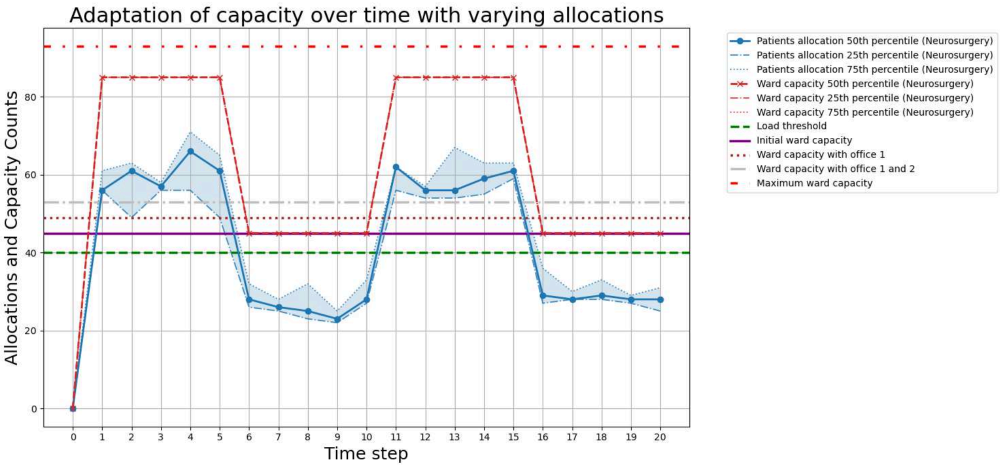

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- Runtime adaptation: opens/closes spare rooms in response to demand.
- Alignment: semantic lifting keeps the DT consistent after human actions.
- Cost awareness: penalties encode policy (offices before corridor)
- Trade-off: correctness/alignment vs runtime overhead
- Current limitation: thresholds and penalties are manually tuned

# Future directions

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- **Learned penalties/thresholds** (data-driven tuning)
- **Live data integration** (connect to ward systems rather than simulator surrogate)
- **Broader scope** (multi-ward, staff constraints, other domains)

# Thanks for the attention

## Questions?

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